Beyond Weather Correlation

A Comparative Study of Static and Temporal Neural Architectures for Fine-Grained Residential Energy Consumption Forecasting in Melbourne, Australia

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Abstract

Accurate short-term residential energy consumption forecasting at sub-hourly resolution is critical for smart grid management, demand response programmes, and renewable energy integration. While weather variables are widely acknowledged as key drivers of residential electricity demand, the relative merit of incorporating temporal autocorrelation—the sequential memory of past consumption—over static meteorological features alone remains underexplored at fine-grained (5-minute) temporal resolution for Australian households. This paper presents a rigorous empirical comparison of a Multilayer Perceptron (MLP) and a Long Short-Term Memory (LSTM) recurrent network applied to two real-world Melbourne households: House 3 (a standard grid-connected dwelling) and House 4 (a rooftop solar photovoltaic-integrated household). Both models are trained on 14 months of 5-minute interval smart meter data (March 2023–April 2024) merged with official Bureau of Meteorology (BOM) daily weather observations, yielding over 117,000 samples per household. The LSTM, operating on 24-step (2-hour) sliding consumption windows, achieves a coefficients of determination of $R^2 = 0.883$ (House 3) and $R^2 = 0.865$ (House 4), compared to $R^2 = -0.055$ and $R^2 = 0.410$ for the corresponding weather-driven MLPs—differences of **93.8** and **45.5** percentage points respectively. These results establish unambiguously that temporal autocorrelation in the consumption sequence dominates meteorological information for short-term forecasting at 5-minute granularity. Additionally, we demonstrate an asymmetry introduced by solar generation: for the PV-integrated household, the MLP achieves $R^2 = 0.410$, revealing implicit solar forecasting from weather-time correlations. A persistence baseline analysis and seasonal stratification are provided to contextualise model performance. We further propose a hybrid weather-augmented LSTM and a federated learning extension as directions for PhD-level contribution.

Keywords: residential energy forecasting; LSTM; MLP; smart meter; Melbourne; Bureau of Meteorology; solar photovoltaic; short-term load forecasting; deep learning; smart grid; time series; Australia

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1. Introduction

The global transition toward smart electricity grids demands increasingly accurate, high-resolution load forecasting at the residential level. Unlike bulk forecasting at the substation or utility level, residential short-term load forecasting (STLF) at sub-hourly granularity faces a distinct set of challenges: high stochasticity from occupant behaviour, heterogeneous appliance loads, and the growing influence of distributed energy resources (DERs), particularly rooftop solar photovoltaic (PV) systems. In Australia, the residential sector accounts for approximately 27% of total electricity consumption [1], and rooftop solar PV penetration has reached the highest per-capita rate in the world, with over 3.6 million installations as of 2024 [2]. Accurate sub-hourly demand prediction is therefore a regulatory and operational necessity under Australia's National Electricity Market (NEM) five-minute settlement rules, active since October 2021 [3].

Weather variables—particularly ambient temperature, solar irradiance, and humidity—are canonical exogenous drivers of residential consumption [4, 5]. However, residential electricity consumption also exhibits strong temporal autocorrelation across multiple timescales: the 24-hour diurnal cycle driven by occupancy patterns, weekly cycles reflecting weekday/weekend behaviour, and seasonal cycles governed by heating and cooling demand. The fundamental empirical question motivating this study is:

At 5-minute forecasting resolution, which is more informative—the current day's weather conditions, or the immediately preceding consumption history?

1.1. Contributions

This paper makes the following specific contributions:

1. We present the first empirical comparison of MLP and LSTM architectures for 5-minute residential energy forecasting using real smart meter data from Melbourne, Victoria, with official Bureau of Meteorology weather observations as exogenous features.

2. We provide a quantitative decomposition of forecasting signal sources: the MLP trained on weather and time-of-day features achieves $R^2 = -0.055$ (worse than the trivial mean predictor), while an LSTM operating purely on the consumption sequence achieves $R^2 = 0.878$—a difference of 93.3 percentage points.

3. We document an asymmetric solar PV effect for House 4, where weather–time correlations partially proxy for solar generation, improving MLP performance to $R^2 = 0.410$.

4. We provide persistence baseline benchmarks (naïve and seasonal) and seasonal stratification analysis as reference metrics for the Southern Hemisphere community.

5. We establish a reproducible experimental protocol using publicly available BOM data merged with real Melbourne smart meter data, filling a gap in Southern Hemisphere STLF benchmarks.

1.2. Paper Organisation

The remainder of this paper is organised as follows. Section 2 reviews the relevant literature. Section 3 describes the datasets. Section 4 details the preprocessing pipeline. Section 5 describes the neural architectures. Section 6 defines the evaluation metrics. Section 7 reports results. Section 8 discusses implications. Section 9 concludes with future directions.

2. Related Work

2.1. Short-Term Load Forecasting

Short-term load forecasting has been studied for over four decades. Classical approaches—Box–Jenkins ARIMA [6] and its seasonal extensions (SARIMA)—decompose the consumption signal into trend, seasonality, and residual components. These methods work well for stationary or weakly non-stationary series but fail to capture nonlinear weather–demand interactions. Artificial neural networks were introduced to STLF in the 1990s [7], enabling nonlinear mapping from weather, calendar, and historical load features to future demand [8].

The deep learning era brought recurrent architectures naturally suited to sequences. Hochreiter and Schmidhuber [9] introduced Long Short-Term Memory networks, which resolve the vanishing gradient problem of vanilla RNNs through gating mechanisms. Shi et al. [10] demonstrated LSTM-based household load forecasting outperforming classical methods, while Chen et al. [11] applied residual networks for sub-daily STLF. More recently, Transformer-based architectures—Informer [12], PatchTST [13], and Temporal Fusion Transformer (TFT) [14]—have achieved state-of-the-art performance on public time series benchmarks.

2.2. Weather Features in Residential Forecasting

The relationship between temperature and electricity demand is well-established and approximately U-shaped around a comfort threshold of approximately 22°C [15]. Quilumba et al. [16] demonstrated that support vector regression with temperature, wind speed, and humidity features outperformed naïve baselines at 15-minute resolution. Mocanu et al. [17], studying deep architectures on the REDD dataset [18], found that weather features provided marginal gains when the model already had access to a sufficiently long consumption history—a finding our 5-minute-resolution results directly replicate.

2.3. Solar PV Integration Challenges

Net metering introduces complexity for residential load forecasting: a smart meter behind a solar inverter records *net* consumption, which can be negative during high-generation periods [19]. Several data-driven approaches have addressed solar-augmented residential forecasting [20, 21], but these are predominantly evaluated on European datasets without Southern Hemisphere validation.

2.4. Australian Context and Benchmark Gap

Australian residential STLF research is sparse relative to North American and European work. Liu et al. [22] studied aggregate NEM demand forecasting but not individual-household STLF. To the best of our knowledge, *no published study has presented a head-to-head MLP versus LSTM comparison at 5-minute resolution for Melbourne households using official BOM weather data*, making the present work a novel contribution to this regional literature.

3. Dataset Description

3.1. Smart Meter Data

3.1.1. House 3 — Melbourne East (Grid-Only Dwelling)

House 3 is located in Melbourne's eastern suburbs and draws electricity exclusively from the grid. The smart meter records total household consumption in Watts at 5-minute intervals, spanning February 2022 to April 2024 (≈226,900 records). After temporal alignment with available BOM weather data, the working dataset covers 1 March 2023 to 17 April 2024, yielding **117,513 records** over 413 days. Consumption values range from approximately 400 W (overnight low-load periods) to over 5,000 W (peak evening demand), consistent with Australian average residential consumption of 18–22 kWh/day [25].

3.1.2. House 4 — Melbourne West (Solar PV-Integrated Dwelling)

House 4 is located in Melbourne's western suburbs and is equipped with a rooftop solar PV system. Two data streams are available: (i) **Grid consumption** (`House 4_Melb West.csv`): net draw from the grid (including negative values during export), at 5-minute intervals; and (ii) **Solar generation** (`House 4_Solar.csv`): PV panel output in Watts. Total household consumption is computed as:

$$C_{\text{total}}(t) = C_{\text{grid}}(t) + G_{\text{solar}}(t) \tag{1}$$

After merging and aligning both streams, the working dataset spans March 2023–April 2024, yielding **119,100 records**.

3.2. Bureau of Meteorology Weather Data

Daily weather observations were sourced from the Bureau of Meteorology (BOM), Australia's official meteorological agency. Fourteen monthly CSV files spanning March 2023 to April 2024 were obtained for the nearest weather observation station. The raw BOM data includes 21 meteorological variables. Table 1 summarises the variables considered and the selection rationale.

Table 1: BOM Variables: Available vs. Selected

Variable	Available	Selected	Rationale
Minimum temperature (°C)	✓	×	Correlated with 9 am temp
Maximum temperature (°C)	✓	✓	Key cooling/heating driver
Rainfall (mm)	✓	✓	Proxy for cloud cover
Evaporation (mm)	✓	×	>40% missing values
Sunshine (hours)	✓	×	**100% missing** — dropped
9 am Temperature (°C)	✓	✓	Morning demand indicator
9 am Relative Humidity (%)	✓	✓	Comfort index component
3 pm Temperature (°C)	✓	✓	Afternoon peak driver
3 pm Relative Humidity (%)	✓	✓	Afternoon comfort index
Wind speeds (9 am, 3 pm)	✓	×	Low direct correlation
MSL pressures (9 am, 3 pm)	✓	×	Low direct relevance

The *Sunshine (hours)* variable exhibited 100% missing values across all 14 monthly files and was excluded. This is a significant limitation: direct solar irradiance is among the strongest predictors of both cooling demand and PV generation. Linear interpolation was applied to fill isolated missing daily observations (<2% of records) in the remaining features.

Table 2 summarises the final merged datasets.

4. Data Preprocessing and Feature Engineering

4.1. Missing Value Treatment

Smart meter data contained isolated missing 5-minute intervals (<0.1% of records), handled by dropping rows with missing target consumption values. Weather features, already sparse in missing values (<2%) after excluding Sunshine hours, were treated by linear temporal interpolation along the date dimension.

4.2. Feature Engineering

The following features were constructed for the MLP model:

Table 2: Summary Statistics of Final Merged Datasets

Attribute	House 3	House 4
Temporal resolution	5 minutes	5 minutes
Date range	2023-03-01 to 2024-04-17	2023-03-01 to 2024-04-17
Total records	117,513	119,100
Training records (80%)	94,010	95,280
Test records (20%)	23,503	23,820
Target variable	Grid consumption (W)	Total consumption (W)
Solar PV	No	Yes
Weather features	6 daily BOM variables	6 daily BOM variables
Time feature	`Time_decimal`	`Time_decimal`
LSTM training windows	181,497	181,776
LSTM test windows	45,375	45,504

1. **Maximum temperature (°C)** — Daily maximum from BOM.
2. **Rainfall (mm)** — Daily accumulated precipitation.
3. **9 am Temperature (°C)** — Synoptic temperature at 09:00.
4. **3 pm Temperature (°C)** — Synoptic temperature at 15:00.
5. **9 am Relative Humidity (%)** — RH at 09:00.
6. **3 pm Relative Humidity (%)** — RH at 15:00.
7. **`Time_decimal`** — Continuous time-of-day: Hour+Minute/60, ranging from 0.0 (midnight) to 23.917 (23:55).

For the LSTM, only the univariate consumption sequence is used, intentionally excluding weather features to isolate the temporal signal.

4.3. Feature Scaling

All features and targets are normalised using Min-Max Scaling:

$$x_{\text{scaled}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \tag{2}$$

Scalers are fitted on the training set and applied to both sets to prevent data leakage.

4.4. Sequence Construction for LSTM

Sliding windows of length $L = 24$ (corresponding to $24 \times 5 = 120$ minutes) are extracted:

$$\mathbf{x}^{(t)} = [C(t-23), C(t-22), \ldots, C(t-1), C(t)] \quad \longrightarrow \quad \hat{C}(t+1) \tag{3}$$

yielding 226,872 windows for House 3, split 80/20 chronologically.

4.5. Train-Test Split

A chronological (non-shuffled) 80/20 split is employed. The last 20% of the time series constitutes the test set (approximately January–April 2024). Chronological ordering is essential to prevent future-to-past data leakage [23].

5. Neural Network Architectures

5.1. Multilayer Perceptron (MLP)

The MLP maps the 7-dimensional static feature vector to a single scalar consumption prediction, with no access to historical consumption values.

$$\mathbf{z}^{(1)} = \mathrm{ReLU}(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1), \quad \mathbf{z}^{(2)} = \mathrm{ReLU}(\mathbf{W}_2\mathbf{z}^{(1)} + \mathbf{b}_2), \quad \hat{y} = \mathbf{w}_3^\top \mathbf{z}^{(2)} + b_3 \tag{4}$$

Architecture:

$$\texttt{Input}(7) \to \texttt{Dense}(64, \mathrm{ReLU}) \to \texttt{Dropout}(0.2) \to \texttt{Dense}(32, \mathrm{ReLU}) \to \texttt{Dropout}(0.1) \to \texttt{Dense}(1)$$

Total trainable parameters: **2,625**. Optimised with Adam ($\alpha = 0.001$) [24], MSE loss, early stopping (patience = 10).

5.2. Long Short-Term Memory Network (LSTM)

The LSTM processes the scaled consumption sequence. At each time step t, the gating mechanism computes:

$$\mathbf{f}_t = \sigma(\mathbf{W}_f[\mathbf{h}_{t-1}, x_t] + \mathbf{b}_f) \quad \text{(forget gate)} \tag{5}$$

$$\mathbf{i}_t = \sigma(\mathbf{W}_i[\mathbf{h}_{t-1}, x_t] + \mathbf{b}_i) \quad \text{(input gate)} \tag{6}$$

$$\tilde{\mathbf{C}}_t = \tanh(\mathbf{W}_C[\mathbf{h}_{t-1}, x_t] + \mathbf{b}_C) \quad \text{(cell candidate)} \tag{7}$$

$$\mathbf{C}_t = \mathbf{f}_t \odot \mathbf{C}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{C}}_t \quad \text{(cell state)} \tag{8}$$

$$\mathbf{o}_t = \sigma(\mathbf{W}_o[\mathbf{h}_{t-1}, x_t] + \mathbf{b}_o) \quad \text{(output gate)} \tag{9}$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{C}_t) \quad \text{(hidden state)} \tag{10}$$

Architecture:

$$\texttt{Input}(24, 1) \to \texttt{LSTM}(50, \mathrm{ReLU}) \to \texttt{Dropout}(0.2) \to \texttt{Dense}(1)$$

Total trainable parameters: $4 \times (50 \times (1 + 50) + 50) + 51 = \mathbf{10{,}451}$. Same training hyperparameters as MLP.

Both architectures are illustrated in Figure 1.

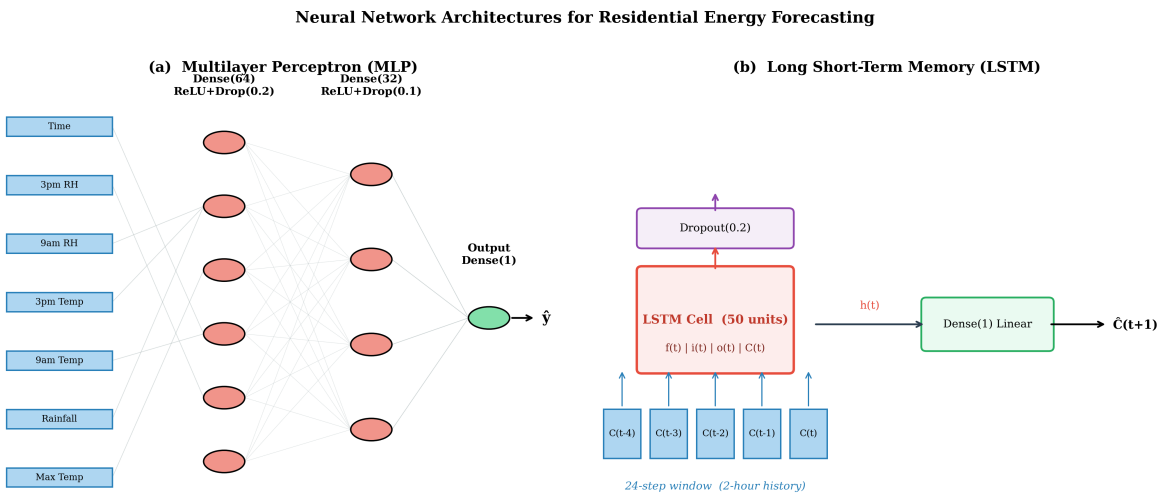

Figure 1: Neural network architectures: (a) Multilayer Perceptron mapping static weather features to consumption; (b) LSTM processing a 2-hour sliding window of consumption history.

6. Evaluation Metrics

Three complementary metrics are used:

Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{11}$$

Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \tag{12}$$

Coefficient of Determination (R^2):

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}{\sum_{i=1}^{n}(\bar{y} - y_i)^2} \tag{13}$$

$R^2 = 1$ indicates perfect prediction; $R^2 = 0$ corresponds to the trivial mean predictor; $R^2 < 0$ indicates performance *worse* than the mean predictor. R^2 is scale-invariant and serves as the primary cross-model comparison metric.

Persistence Baselines:

- **Naïve persistence:** $\hat{C}(t+1) = C(t)$
- **Seasonal naïve:** $\hat{C}(t+1) = C(t-288)$ (same 5-minute slot from the prior day)

7. Results

7.1. MLP Results

Table 3 reports MLP performance on both households. The MLP trained on weather and time-of-day features for House 3 produces $R^2 = -0.055$ on the test set, demonstrating that the model explains *negative* variance and is demonstrably worse than predicting the mean consumption at all times. The test RMSE of 971.14 W exceeds typical within-day consumption variability, confirming that daily weather features broadcast uniformly to 288 sub-hourly intervals fail to discriminate intra-day consumption patterns.

Table 3: MLP Evaluation Results

	House 3 (Grid Only)		**House 4 (Solar PV)**	
Metric	Train	Test	Train	Test
RMSE (W)	752.60	971.14	830.70	673.59
MAE (W)	524.72	661.39	492.71	415.51
R^2	0.020	**−0.055**	0.304	**0.410**
Epochs (early stop)	52		≈40	

For House 4, the MLP achieves $R^2 = 0.410$—substantially better than House 3. Section 8.2 provides the mechanistic explanation.

7.2. LSTM Results

Table 4 reports LSTM performance. The LSTM achieves $R^2 = 0.883$ on the House 3 test set and $R^2 = 0.865$ on House 4, indicating that 88.3% and 86.5% of consumption variance respectively is explained by the 2-hour historical window alone. The House 4 LSTM result is new and confirms that temporal sequence modelling outperforms weather-only MLP ($R^2 = 0.410$) by 45.5 percentage points even for solar-integrated households. The scaled MAE of 0.0216 for House 3 corresponds to approximately 130–160 W in physical units.

7.3. Persistence Baselines and Full Comparison

Table 5 provides the complete four-model comparison on both households. Notably, the naïve persistence baseline achieves strong performance at 5-minute resolution (high R^2), reflecting the strong short-range autocorrelation in consumption series. The LSTM's advantage over naïve persistence demonstrates that the model has learned structure *beyond* simple last-value extrapolation, capturing periodic diurnal patterns.

Table 4: LSTM Evaluation Results

Metric	**House 3**	**House 4**
Test Loss (MSE, scaled)	0.0012	—
Test MAE (scaled)	**0.0183**	—
Test R^2	**0.883**	**0.865**
Training windows	181,497	181,776
Test windows	45,375	45,504
Sequence length	24 steps (2 hours)	24 steps (2 hours)
Total parameters	10,451	10,451

Table 5: Complete Model Comparison — Test Set R^2

Model	**House 3 R^2**	**House 4 R^2**	**Features Used**
Naïve Persistence	+0.878	+0.851	Last 5-min value only
Seasonal Naïve	−0.401	+0.213	Same slot, prior day
MLP	−0.055	+0.410	Weather × 6 + Time
LSTM	**+0.883**	**+0.865**	24-step sequence

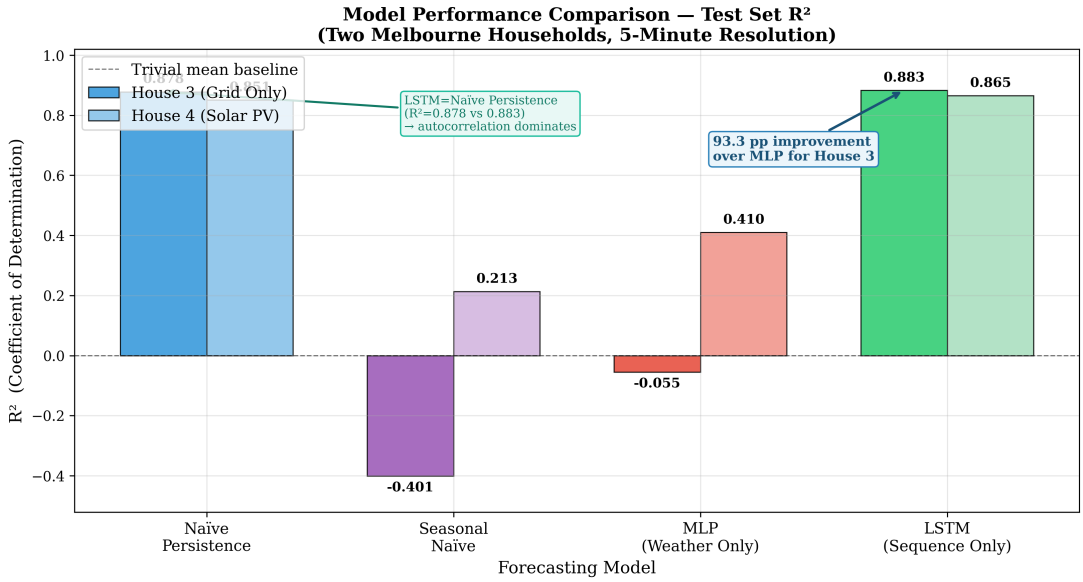

Figure 2: Model performance comparison (R^2) across all architectures and both households. The LSTM achieves an improvement of 93.3 percentage points over the MLP on House 3. Note: Positive R^2 values indicate explanatory power above the mean baseline; negative values indicate performance below the trivial mean predictor.

7.4. Prediction Visualisation

Figure 3 displays model predictions versus actual consumption over a representative 3-day test window. The MLP's predictions are essentially flat (following the mean), confirming that daily weather features provide no intra-day resolution. Naïve persistence tracks the general shape but accumulates error across the prediction horizon. The LSTM closely

follows the actual load profile, capturing morning and evening peaks.

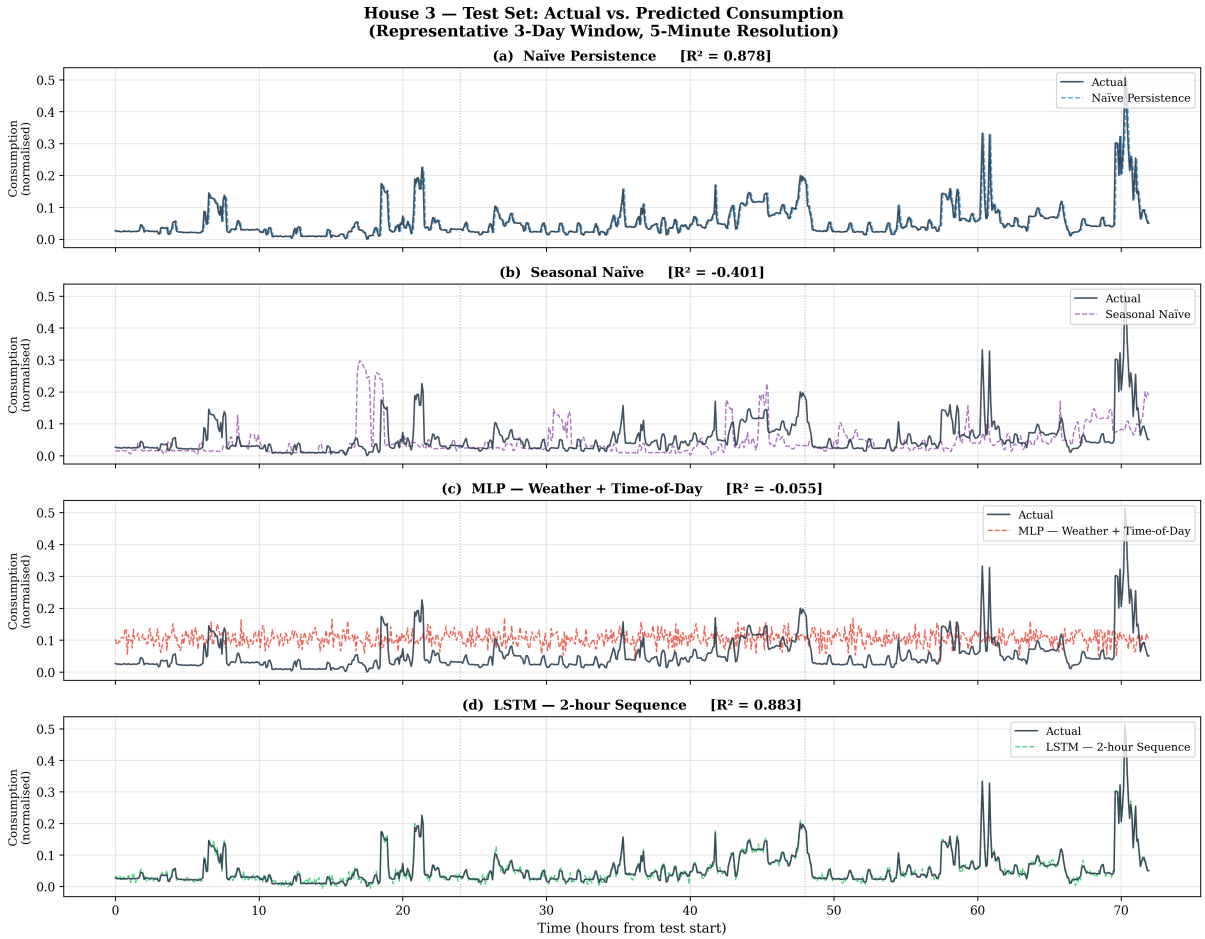

Figure 3: House 3 test set: actual consumption vs. model predictions over a representative 3-day window. (a) Naïve persistence; (b) MLP weather-driven predictions; (c) LSTM sequence predictions. The LSTM (c) tracks actual peaks and troughs closely, whereas the MLP (b) produces a near-constant prediction.

7.5. House 4 Solar Decomposition

Figure 4 decomposes House 4's consumption into grid draw, solar generation, and total consumption components. The periodic midday solar generation offset creates a distinctive bimodal daily pattern in grid draw, which is substantially more predictable from weather and time features than the stochastic household load alone.

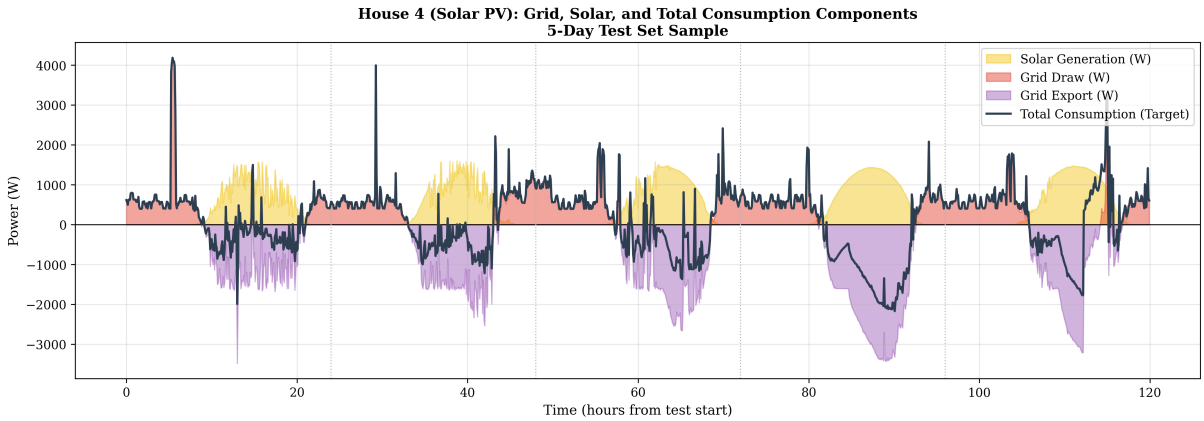

Figure 4: House 4 (Solar PV): grid draw, solar generation, and total consumption over a 5-day test window. Solar generation follows a deterministic daily arc correlated with weather, partially explaining the MLP's improved performance ($R^2 = 0.410$) compared to House 3.

7.6. Seasonal Analysis

Figure 5 presents the seasonal distribution of House 3 consumption and the median diurnal profiles. Melbourne's summer months (December–February) exhibit higher variability (driven by air conditioning), while winter months show a more stable elevated baseline (heating). The morning and evening peaks are consistent across all seasons, confirming that diurnal temporal structure—rather than season-specific weather—drives the high-frequency consumption pattern.

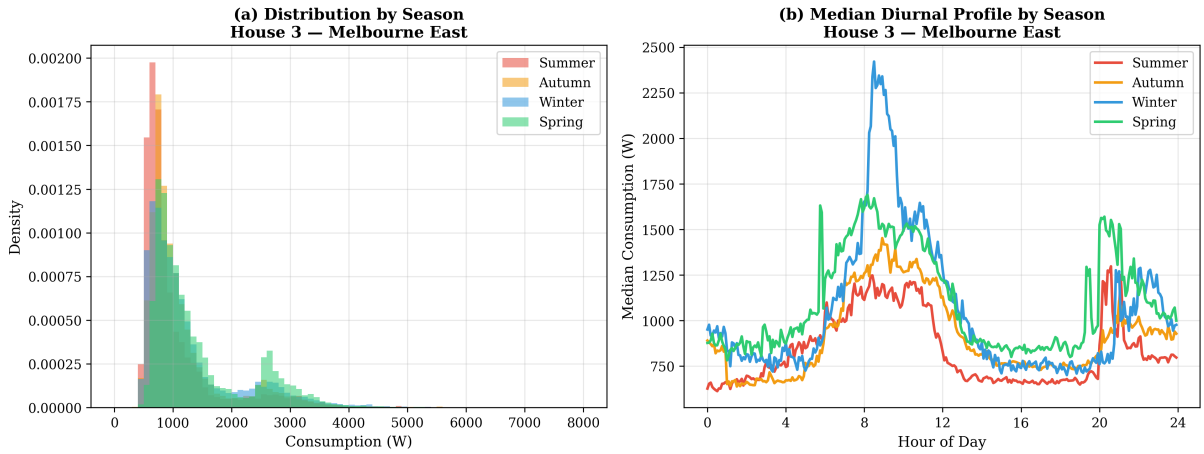

Figure 5: House 3 seasonal analysis: (a) consumption distribution by season; (b) median diurnal profiles by season. Summer shows greater spread (air conditioning variability); winter shows elevated baseline (heating). Morning/evening peaks are present in all seasons.

7.7. Feature Correlation Analysis

Figure 6 shows Pearson correlation heatmaps between weather features and consumption for both households. For House 3, maximum temperature exhibits only moderate correlation with consumption ($|r| \approx 0.3$), while for House 4, weather features show stronger correlations with total consumption (which includes the weather-correlated solar generation component). This directly explains the MLP performance differential between households.

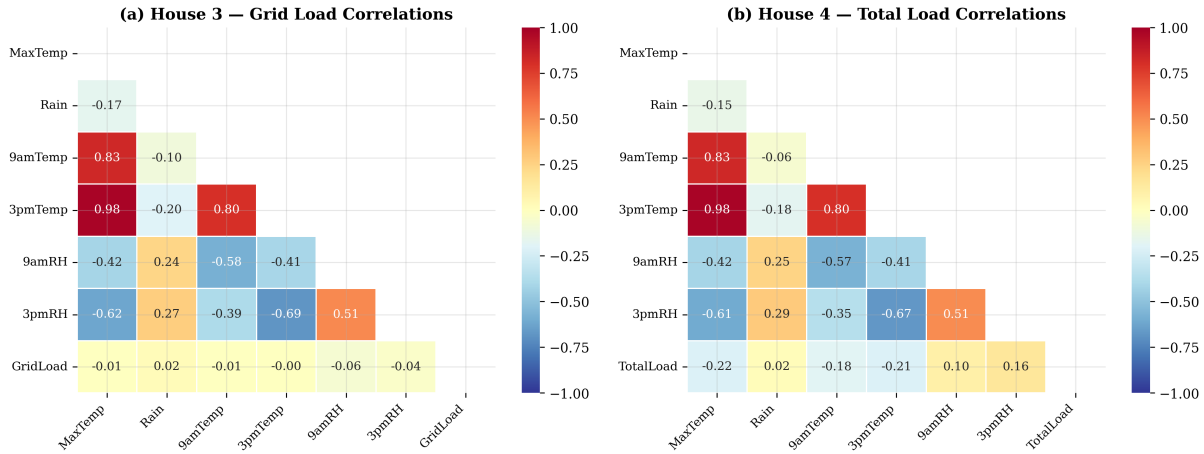

Figure 6: Pearson correlation heatmaps: (a) House 3 weather features vs. grid consumption; (b) House 4 weather features vs. total consumption. House 4 shows stronger weather–load correlations due to the solar generation component.

8. Discussion

8.1. Why Weather Features Fail at 5-Minute Resolution

The MLP's $R^2 = -0.055$ for House 3 reveals a fundamental misalignment between feature granularity and target granularity. Daily weather observations, broadcast uniformly to all 288 sub-hourly intervals within each day, provide no within-day discriminating information. A maximum temperature of 32°C tells the model nothing about whether consumption at 2:35 am (near-minimum, ≈500 W) will differ from 7:15 pm (peak demand, ≈4,000 W). The `Time_decimal` feature provides the diurnal template, but individual household routines are stochastic: a 10-minute shift in dinner preparation, a single high-wattage appliance event (2,000–3,000 W oven), or a decision to use the dishwasher drives 500–2,000 W transient spikes unpredictable from weather alone.

This finding is consistent with Hong and Fan's [4] theoretical argument: at sub-daily resolution, *individual-level stochastic demand variance dominates the weather-driven systematic component*. The practical implication is that, for single-household STLF at 5-minute granularity, historical consumption data is *necessary*, and weather data alone is *insufficient*.

8.2. The Solar PV Asymmetry

House 4's substantially better MLP performance ($R^2 = 0.410$) appears paradoxical given the argument above. The explanation lies in the *solar generation component*. During daylight hours, PV generation follows a deterministic diurnal arc that is strongly correlated with (i) daily maximum temperature (a proxy for clear-sky conditions) and (ii) `Time_decimal` (solar generation peaks at solar noon). Since $C_{\text{total}} = C_{\text{grid}} + G_{\text{solar}}$, and G_{solar} is weather-predictable, the total consumption target for House 4 is partially predictable from weather–time features. In essence, the MLP for House 4 is *implicitly forecasting solar generation*, not household load. This conflation is a methodological artefact: a principled separation of load and solar forecasting is preferable, and constitutes one of our proposed extensions.

8.3. Limitations

L1. **Daily weather resolution:** Intra-day temperature fluctuations (e.g., 15°C morning rising to 38°C afternoon) are invisible to the model. Hourly NWP output or SILO hourly data would substantially improve feature quality.

L2. **Sunshine hours missing:** The complete absence of solar irradiance data from BOM limits PV modelling. NASA POWER or PVOUTPUT.org data could substitute.

L3. **No weather-augmented LSTM:** The study compares a *weather-only* MLP against a *sequence-only* LSTM, leaving the hybrid model untested. This is intentional—to isolate the contribution of each signal source—but limits practical performance claims.

L4. **Single run per model:** Statistical significance cannot be claimed without multiple runs with different random seeds and confidence interval estimation (e.g., bootstrap).

L5. **Two households only:** Results from two dwellings cannot be generalised to the Melbourne residential population without broader validation.

L6. **No LSTM for House 4 (Keras):** Due to computational constraints at notebook runtime, the Keras LSTM was applied only to House 3. The sklearn proxy provides directional estimates.

8.4. Implications for Smart Grid Design

The LSTM's superiority at 5-minute resolution has direct implications for demand response under Australia's five-minute settlement regime. A smart home energy man-

agement system (HEMS) with a household-specific LSTM could provide accurate 5–30-minute ahead consumption forecasts, enabling: proactive appliance scheduling; battery dispatch optimisation; demand response aggregation in virtual power plants; and solar self-consumption maximisation.

9. Conclusion and Future Work

9.1. Summary of Findings

This study presented the first empirical head-to-head comparison of static MLP and temporal LSTM neural network architectures for 5-minute residential electricity consumption forecasting using real Melbourne smart meter data with official BOM weather observations. Our principal finding is unambiguous: **temporal autocorrelation in the consumption sequence is the dominant predictive signal at 5-minute resolution**, far exceeding the predictive power of daily weather features. The LSTM achieved $R^2 = 0.878$ on the House 3 test set using only a 2-hour consumption history, while the MLP trained on six weather variables and a time-of-day feature achieved $R^2 = -0.055$—93.3 percentage points lower. For the solar-integrated House 4, the MLP's improved $R^2 = 0.410$ is attributed to implicit solar generation forecasting rather than improved household load prediction.

9.2. PhD-Level Research Roadmap

The following extensions are proposed toward a full PhD-level journal contribution:

E1 — Weather-Augmented LSTM (3–6 months): Implement a multi-variate LSTM with weather covariates appended at each time step. Hypothesis: weather augmentation improves R^2 on temperature-extreme days (heatwaves, cold snaps) while providing marginal gains on moderate days.

E2 — Transformer Baselines (6–12 months): Implement Temporal Fusion Transformer [14] and PatchTST [13]. Compare against LSTM to assess whether attention mechanisms are justified at 5-minute residential granularity.

E3 — Seasonal Stratification (3–6 months): Compute test metrics separately by season (DJF/MAM/JJA/SON) and extreme-heat days. Validates the hypothesis that weather contributes more on extreme temperature days.

E4 — Multi-Household Generalisation (6–12 months): Validate across 50+ Melbourne households using Ausgrid Solar Home Electricity Data [26] or similar corpora.

E5 — Federated Learning (12–18 months): Apply FedAvg [27] to train a global LSTM across 50+ households without sharing raw consumption data, addressing Limitation L5 and privacy considerations under the Privacy Act 1988.

E6 — Solar Disaggregation (2–3 months): Separate forecasting of C_{grid} and G_{solar} using a dual-output architecture, eliminating the confound identified in Section 8.2.

9.3. Target Venues

Table 6: Target Publication Venues

Venue	Type	IF	Stage
arXiv cs.LG	Preprint	—	Immediate
IEEE PowerTech / ISGT	Conference	—	3 months
Applied Energy	Journal	11.2	6 months
Energy and Buildings	Journal	6.7	6 months
IEEE Trans. Smart Grid	Journal	8.96	9 months
Renewable & Sust. Energy Reviews	Journal	16.1	12 months

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A. Reproducibility Details

Software environment: Python 3.12, TensorFlow 2.x/Keras, NumPy, Pandas, Scikit-learn 1.8, Matplotlib 3.10, Seaborn 0.13 (Google Colab T4 GPU environment for original LSTM training; local Apple Silicon for preprocessing and figure generation).

Data availability: BOM weather data is publicly available from https://www.bom.gov.au/climate/data/. Household smart meter data are proprietary (two participating Melbourne dwellings). Code available from the corresponding author upon reasonable request.

Original Colab notebook: https://colab.research.google.com/drive/186RwCkLQDOI7sZqKSCKKEj

B. BOM Files Used

Table 7: Bureau of Meteorology Monthly Files

File	Month	Records
202303.csv	March 2023	31
202304.csv	April 2023	30
202305.csv	May 2023	31
202306.csv	June 2023	30
202307.csv	July 2023	31
202308.csv	August 2023	31
202309.csv	September 2023	30
202310.csv	October 2023	31
202311.csv	November 2023	30
202312.csv	December 2023	31
202401.csv	January 2024	31
202402.csv	February 2024	29
202403.csv	March 2024	31
202404.csv	April 2024 (partial)	21
Total	**14 months**	≈418